



Calculation Sheet

Calculation header

Identifier

*New calculation (1)***Medium selection and state**

Medium

M4

State

*Liquid***Operating data**☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	<i>100.0</i>	<i>100.0</i>	<i>0.0</i>	°C
p1	<i>3.0</i>	<i>3.0</i>	<i>0.0</i>	kgf/cm ² (g)
○ p2	<i>2.3</i>	<i>2.3</i>		kgf/cm ² (g)
● Δp	<i>0.7</i>	<i>0.7</i>	✗ <i>0.0</i>	kgf/cm ²
Cv	<i>15.125</i>	<i>15.125</i>		GPM(US)
● qm	<i>11,750.0</i>	<i>11,750.0</i>	✗ <i>0.0</i>	kg/h
○ qv	<i>10.0</i>	<i>10.0</i>		m ³ /h
s/s100	<i>71.568</i>	<i>71.568</i>		%
LpAe	<i>37.6</i>	<i>37.6</i>		dB(A)

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ρ1	<i>1,175.0</i>	<i>1,175.0</i>	✗ <i>0.0</i>	kg/m ³
ρ2	<i>1,175.0</i>	<i>1,175.0</i>		kg/m ³
● η1	✗ <i>0.0</i>	✗ <i>0.0</i>	✗ <i>0.0</i>	mPa s
cF1	<i>0.0</i>	<i>0.0</i>	<i>0.0</i>	m/s

Pipe downstream of valve

Size class downstream of valve

NPS2

2"

Schedule downstream

SCH2

*STD***Valve configuration**

Valve type

Straight globe valve

Trim type

Cage trim



Calculation Sheet

Valve configuration (continued)

Flow direction	<i>FTO</i>
Valve performance class	<i>Heavy duty valve</i>
Protection	<i>Hardened seat/plug</i>
Low-noise design	<i>n.a.</i>

Valve data

Basic characteristic		<i>Equal percentage</i>	
Nominal flow coefficient	Cv100	<i>46.0</i>	GPM(US)
☐ Lift stopper			
Selected valve size	NPS	<i>2"</i>	
Pressure class	class	<i>class 150</i>	
Maximum operating pressure	p,max	<i>21.0</i>	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL ²	<i>0.814</i>	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	<i>0.734</i>	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	<i>0.326</i>	-
Valve style modifier (Cv/Cv100 = 1)	Fd	<i>0.41</i>	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
s/s100	<i>71.568</i>	<i>71.568</i>		%
u2	<i>1.3705</i>	<i>1.3705</i>		m/s
LpAe	<i>37.6</i>	<i>37.6</i>		dB(A)

Noise calculation

Calculation standard (noise, liquid)	<i>IEC 60534-8-4 (2015)</i>
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Noise prediction data

Low-noise design		<i>n.a.</i>	
Sound velocity	cF1	<i>0.0</i>	m/s
Pressure level	LA1	<i>139.0</i>	dB
Sound pressure level	Lpi	<i>139.0</i>	dB
Transmission loss	TL	<i>-88.0</i>	dB
Sound power level (A-weighted)	LwAe	<i>48.3</i>	dB(A)





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Noise prediction data (continued)

Sound pressure level of valve (A-weighted)	LpAe	37.6	dB(A)
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Legend

 Errors